

Midterm: instructor key and worked solutions

PRIVATE: keep separate from student booklets.

Each form has 10 medium questions (1–10) and 10 hard questions (11–20), one point each. The forms use equivalent learning objectives, matched arithmetic, and different answer placements. Numerical variants change stated inputs; symbolic questions keep the same logical task. Difficulty and 90-minute timing are design judgments, not student-piloted results.

Question	M7KD	M3RX	M8VN	M2HF	M6WP
1	G	E	C	F	B
2	C	G	B	D	A
3	F	B	A	E	G
4	C	F	G	A	E
5	C	E	F	G	D
6	E	A	B	D	C
7	D	A	E	B	F
8	D	B	F	A	G
9	A	D	G	E	C
10	F	G	D	A	B
11	G	C	E	B	D
12	D	E	C	G	F
13	B	D	C	E	A
14	A	F	D	G	E
15	E	C	F	B	A
16	G	C	A	F	B
17	B	D	G	C	E
18	B	A	E	C	F
19	A	F	B	D	G
20	E	G	A	F	D

Scope and teaching-source map

The full Lecture 1 and Lecture 2 slide sources define the boundary. Practice Sets 1–4 and workshop Questions 1–9 informed the design. The separate Lecture 3 deck, later workshop questions, coefficient tests, and confidence-interval construction are not needed. All probabilities and numerical examples are invented; no external data or student records are used. The first nine hard items map respectively to the nine questions worked in class; question 20 integrates OVB and large-sample reasoning already present in Lecture 2 and Practice Set 4.

1. **Medium: Conditioning and group averages.** slides/lecture-01.qmd: Conditioning changes the reference group; Conditional mean; The law of iterated expectations; slides/practice-01.qmd: Questions 8-9; slides/lecture-05.qmd: Q1.
2. **Medium: Moments after a change of units.** slides/lecture-01.qmd: Mean; Variance; What happens when units change?; slides/practice-01.qmd: Questions 6-7.
3. **Medium: Rescaling covariance and correlation.** slides/lecture-01.qmd: Covariance; Correlation; slides/practice-01.qmd: Question 10.
4. **Medium: Efficiency of fixed weights.** slides/lecture-02.qmd: Unequal weights; BLUE; slides/practice-02.qmd: Questions 8 and 13.
5. **Medium: Bias and consistency.** slides/lecture-02.qmd: Unbiasedness and consistency are different; slides/practice-02.qmd: Question 7.
6. **Medium: Sample variance and standard error.** slides/lecture-02.qmd: Sample variance; Standard deviation and standard error; slides/practice-02.qmd: Question 6.
7. **Medium: A two-sided test of a mean.** slides/lecture-02.qmd: Testing a claim about the mean; slides/practice-02.qmd: Questions 5 and 10.
8. **Medium: OLS coefficients from summaries.** slides/lecture-02.qmd: Center the data to solve for the slope; slides/practice-03.qmd: Questions 7-8.
9. **Medium: Fit and residual scale.** slides/lecture-02.qmd: Measures of fit and prediction accuracy; slides/practice-03.qmd: Questions 6 and 9-10.
10. **Medium: Omission and the probability limit.** slides/lecture-02.qmd: Derive the omitted-variable distortion; LLN; slides/practice-04.qmd: Questions 4, 9-10.
11. **Hard: Missing joint cells and reverse conditioning.** slides/lecture-01.qmd: Joint and marginal distributions; Conditioning; slides/practice-01.qmd: Question 8; slides/lecture-05.qmd: Q1-Q2.
12. **Hard: Repair an iterated-expectations proof.** slides/lecture-01.qmd: The law of iterated expectations, proof; slides/lecture-05.qmd: Q2.
13. **Hard: Zero covariance with a nonlinear error.** slides/lecture-01.qmd: Zero correlation can hide a relationship; slides/lecture-02.qmd: Zero covariance is not the full conditional-mean assumption; slides/practice-03.qmd: Question 23; slides/lecture-05.qmd: Q3.
14. **Hard: Weights, concentration, and relative variance.** slides/lecture-02.qmd: Unequal weights; BLUE; Consistency; slides/practice-02.qmd: Questions 7-8,13; slides/lecture-05.qmd: Q4.
15. **Hard: Outcome-dependent weights.** slides/lecture-01.qmd: Bernoulli distribution; Independence; slides/lecture-02.qmd: BLUE fixed weights; slides/lecture-05.qmd: Q5.
16. **Hard: Vanishing bias at the CLT scale.** slides/lecture-02.qmd: Unbiasedness versus consistency; LLN/CLT; Slutsky; slides/practice-02.qmd: Questions 7,14; slides/lecture-05.qmd: Q6.
17. **Hard: Duplicated observations and testing.** slides/lecture-01.qmd: i.i.d.; What averaging buys us; slides/lecture-02.qmd: Known-variance mean test; Two-sided rejection; slides/lecture-05.qmd: Q7.
18. **Hard: Recover outcomes and fit from normal equations.** slides/lecture-02.qmd: Normal equations; Fit decomposition; slides/practice-03.qmd: Questions 8-9,16-17; slides/lecture-05.qmd: Q8.
19. **Hard: Complete the fit-decomposition argument.** slides/lecture-02.qmd: Why total variation decomposes exactly; Zero residual correlation is not evidence of exogeneity; slides/practice-03.qmd: Questions 16-17; slides/lecture-05.qmd: Q9.
20. **Hard: Confounding and concentration around the wrong target.** slides/lecture-02.qmd: OVB; LLN/CLT and OLS sampling distribution; slides/practice-04.qmd: Questions 9,13,15.

Form M7KD: worked solutions

1. Conditioning and group averages (medium)

Correct answer: G. $(\frac{4}{3}, 1)$

$\Pr(X = 1) = 0.60$, so $E(Y | X = 1) = 2(0.40/0.60) = \frac{4}{3}$. Marginally $\Pr(Y = 2) = 0.50$, hence $E(Y) = 2(0.50) = 1$. Equivalently, average the two conditional means with weights 0.40 and 0.60, not equal weights. These are descriptive population quantities.

2. Moments after a change of units (medium)

Correct answer: C. $(3, 8)$

$E(Y) = 2$ and $E(Y^2) = 6$, so $\text{Var}(Y) = 6 - 2^2 = 2$. Therefore $E(W) = 7 - 2(2) = 3$ and $\text{Var}(W) = (-2)^2(2) = 8$. The intercept shifts the mean only; variance uses the squared scale.

3. Rescaling covariance and correlation (medium)

Correct answer: F. $(24, \frac{1}{2})$

$\text{Cov}(X, Y) = (-1/2)(2)(4) = -4$. Thus $\text{Cov}(W, V) = 2(-3)(-4) = 24$. The single negative scale reverses the correlation sign, giving $+1/2$; location shifts do not affect covariance or correlation.

4. Efficiency of fixed weights (medium)

Correct answer: C. $(\frac{3}{2}, \frac{9}{8})$

Both sets of fixed weights sum to one, so both rules are unbiased. Independence and squared weights give $\text{Var}(B) = 4(1/4 + 1/16 + 1/16) = \frac{3}{2}$, while $\text{Var}(A) = \frac{4}{3}$. The ratio is $9/8$: B has 12.5% more variance, not 12.5% more SD.

5. Bias and consistency (medium)

Correct answer: C. A is unbiased and inconsistent; B is biased and consistent; C is biased and inconsistent.

$E(A_n) = \mu$ but its nondegenerate distribution never changes. $E(B_n) = \mu + 2/n$, so it is biased for each finite n but converges in probability to μ by the LLN. $C_n \rightarrow_p \mu + 2$ and has constant bias 2. Nonvanishing variance alone is not a general proof of inconsistency; the unchanged distribution proves it for A.

6. Sample variance and standard error (medium)

Correct answer: E. $(\frac{20}{3}, \frac{5}{3})$

The sample mean is 6; squared deviations total $9 + 1 + 1 + 9 = 20$. Hence $s_Y^2 = 20/3$ and $SE^2 = s_Y^2/4 = 5/3$. These calculations do not justify an exact Student t reference without normality or a normal approximation with just four observations.

7. A two-sided test of a mean (medium)

Correct answer: D. $t = 2$, $p \approx 0.0455$; reject.

$SE = 12/\sqrt{144} = 1$; $t = (22 - 20)/1 = 2$. Both tails count: $p \approx 2[1 - \Phi(2)] = 0.0455 < 0.05$, so reject. Finite positive variance supplies the usual CLT/Slutsky conditions for the approximation. A p-value is not the probability the null is true. At 1% the same result would not reject.

8. OLS coefficients from summaries (medium)**Correct answer: D.** (3, 2)

$\hat{\beta}_1 = S_{XY}/S_{XX} = 2$ and $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1\bar{X} = 9 - 2(3) = 3$. The line passes through the means. These algebraic facts need no causal assumption.

9. Fit and residual scale (medium)**Correct answer: A.** ($\frac{3}{4}$, 2)

$R^2 = 1 - SSR/TSS = 1 - 1/4 = 3/4$. $SER = \sqrt{SSR/(n-2)} = \sqrt{64/16} = 2$ in outcome units. R-squared is unitless and establishes neither causality nor out-of-sample accuracy; SER is not the slope's SE.

10. Omission and the probability limit (medium)**Correct answer: F.** (2, 2)

The auxiliary slope is $\pi = -2/4 = -1/2$. The distortion is $2\pi = -1$, so $b_s = 3 - 1 = 2$. The LLN sends the sample covariance and variance to their population counterparts, hence $\hat{b}_s \rightarrow_p b_s = 2$, not the causal target 3. More observations do not remove the omitted pathway.

11. Missing joint cells and reverse conditioning (hard)**Correct answer: G.** ($\frac{2}{5}$, $\frac{1}{5}$, $\frac{2}{3}$)

Since Y is 0 or 3, $\Pr(Y = 3) = E(Y)/3 = 0.30$. Thus $b = 0.30 - 0.10 = 0.20$, and $a = 0.60 - b = 0.40$. Reverse conditioning divides by the Y marginal: $\Pr(X = 1 \mid Y = 3) = b/\Pr(Y = 3) = \frac{2}{3}$. Dividing by 0.60 answers the opposite conditioning question.

12. Repair an iterated-expectations proof (hard)**Correct answer: D.** (p_x, p_{xy}); independence is unnecessary.

The outer expectation weights each X group by p_x . The multiplication rule gives $\Pr(Y = y \mid X = x)p_x = (p_{xy}/p_x)p_x = p_{xy}$ for each pair. Then $\sum_x p_{xy} = p_y$, completing the proof. Independence is not used. A product of marginals may give the same final summed expectation, but it is not a valid termwise replacement for a dependent joint distribution.

13. Zero covariance with a nonlinear error (hard)**Correct answer: B.** (0 , $\frac{25}{4}$); zero conditional mean fails.

Symmetry gives $E(X) = E(X^3) = 0$, so $E(Xu) = 0$; also $E(u) = 0$ by centering. But $E(X^2u) = E(X^4) - [E(X^2)]^2 = \text{Var}(X^2) = (3^2 - 2^2)^2/4 = \frac{25}{4} > 0$. Since u is determined by X , $E(u \mid X) = X^2 - \frac{13}{2}$ takes the nonzero values $\pm \frac{5}{2}$. Zero covariance does not imply zero conditional mean or independence.

14. Weights, concentration, and relative variance (hard)**Correct answer: A.** ($\frac{1}{4}$, 5)

Y_1 and $\bar{Y}_{-1}$ are independent, with variances σ^2 and $\sigma^2/(n-1)$. Hence $\text{Var}(T_n)/\sigma^2 = w^2 + (1-w)^2/(n-1) \rightarrow w^2 = \frac{1}{4}$. For S, divide $\sigma^2[k^2/n + (1-k/\sqrt{n})^2/(n-1)]$ by σ^2/n : $k^2 + \frac{n}{n-1}(1-k/\sqrt{n})^2 \rightarrow k^2 + 1 = 5$. Both are unbiased. T converges in probability to $wY_1 + (1-w)\mu$, so is inconsistent; S is consistent but pays a first-order variance cost.

15. Outcome-dependent weights (hard)

Correct answer: E. 0.36; step (2), because weights depend on outcomes.

This rule selects $T = \max(Y_1, Y_2)$, which is zero only if both outcomes are zero. Thus $E(T) = 1 - (1 - p)^2 = 2p - p^2 = 0.36$, with upward bias $p(1 - p) = 0.16$. Step (1) is linearity and is valid without independence. Step (2) incorrectly factors dependent products: independence between the Y observations does not make a selected weight independent of its own Y . Step (3) is valid algebra on the expression immediately before it.

16. Vanishing bias at the CLT scale (hard)

Correct answer: G. $(\mu, N(-1, 1))$

The LLN and vanishing deterministic shift give $B_n \rightarrow_p \mu$. Yet $\sqrt{n}(B_n - \mu)/\sigma = \sqrt{n}(\bar{Y}_n - \mu)/\sigma - 1 \rightarrow_d N(-1, 1)$. The finite-sample bias is $-1\sigma/\sqrt{n}$: it vanishes in levels but remains 1 sampling SDs below zero. A deterministic shift does not add variance.

17. Duplicated observations and testing (hard)

Correct answer: B. $z = 2$, $p \approx 0.0455$; do not reject.

The copied mean equals $4\sum_i Y_i/(4n) = \bar{Y}$ exactly. All four copies of each observation are perfectly dependent, so its sampling SD stays $10/\sqrt{100} = 1$. Therefore $z = (12 - 10)/1 = 2$ and $p = 2(1 - 0.97725) = 0.0455 > 0.01$: do not reject. Treating rows as independent would incorrectly halve the SE and report $z=4$. The normal reference is exact here under the stated normal, known-variance model; CDF values are rounded.

18. Recover outcomes and fit from normal equations (hard)

Correct answer: B. $(3, 5, \frac{5}{9})$

The endpoint residuals are both 2. Write $\hat{u} = (2, a_* - (5), b_* - (7), 2)$. The two normal equations imply $a_* + b_* = 8$ and $a_* + 2b_* = 13$. Subtraction gives $b_* = 5$ and $a_* = 3$; residuals are $2(1, -1, -1, 1)$. Fitted deviations are $2(-3/2, -1/2, 1/2, 3/2)$, so $ESS = 5(2^2)$ and $SSR = 4(2^2)$. By the intercept-OLS decomposition $TSS = 9(2^2)$, hence $R^2 = 5/9$. Missing outcomes need not equal fitted values.

19. Complete the fit-decomposition argument (hard)

Correct answer: A. $\hat{\beta}_1[\sum_i X_i \hat{u}_i - \bar{X} \sum_i \hat{u}_i]$; both sample normal equations.

The intercept equation gives $\hat{Y}_i - \bar{Y} = \hat{\beta}_1(X_i - \bar{X})$. Substitution yields $C = \hat{\beta}_1[\sum_i X_i \hat{u}_i - \bar{X} \sum_i \hat{u}_i]$. Both sums vanish by the fitted normal equations. Thus $TSS = ESS + SSR$. No population exogeneity, independence, normality, or causal interpretation is needed for this sample algebra. Option using only $\hat{\beta}_1 \sum X_i \hat{u}_i$ has the same zero value at the fit, but its claim that population zero conditional mean is required is false.

20. Confounding and concentration around the wrong target (hard)

Correct answer: E. $(\frac{1}{2}, \approx 0.1573, 0)$

$\text{Cov}(X, Z) = (1/2)(2)(2) = 2$, so $\pi = 2/4 = 1/2$ and the omitted distortion is $1/2$. Hence $\beta = b_s - 1/2 = \frac{1}{2}$. At $n=400$ the approximate sampling SD is $5/20 = 1/4$. Relative to the causal target, $\hat{b}_s - \beta$ is approximately $N(1/2, 1/16)$. Bounds $[-1/4, 1/4]$ standardize to $[-3, -1]$: probability $\approx \Phi(-1) - \Phi(-3) = \Phi(3) - \Phi(1) = 0.15731$. The LLN sends $\hat{b}_s$ to $b_s = \beta + 1/2$, outside this fixed tolerance, so the probability tends to zero. The CLT describes shrinking variation around the short slope, not the causal slope.

Form M3RX: worked solutions

1. Conditioning and group averages (medium)

Correct answer: E. $(2, \frac{3}{2})$

$\Pr(X = 1) = 0.60$, so $E(Y | X = 1) = 3(0.40/0.60) = 2$. Marginally $\Pr(Y = 3) = 0.50$, hence $E(Y) = 3(0.50) = \frac{3}{2}$. Equivalently, average the two conditional means with weights 0.40 and 0.60, not equal weights. These are descriptive population quantities.

2. Moments after a change of units (medium)

Correct answer: G. $(4, 18)$

$E(Y) = 2$ and $E(Y^2) = 6$, so $\text{Var}(Y) = 6 - 2^2 = 2$. Therefore $E(W) = 10 - 3(2) = 4$ and $\text{Var}(W) = (-3)^2(2) = 18$. The intercept shifts the mean only; variance uses the squared scale.

3. Rescaling covariance and correlation (medium)

Correct answer: B. $(30, \frac{1}{2})$

$\text{Cov}(X, Y) = (-1/2)(2)(5) = -5$. Thus $\text{Cov}(W, V) = 2(-3)(-5) = 30$. The single negative scale reverses the correlation sign, giving $+1/2$; location shifts do not affect covariance or correlation.

4. Efficiency of fixed weights (medium)

Correct answer: F. $(\frac{27}{8}, \frac{9}{8})$

Both sets of fixed weights sum to one, so both rules are unbiased. Independence and squared weights give $\text{Var}(B) = 9(1/4 + 1/16 + 1/16) = \frac{27}{8}$, while $\text{Var}(A) = 3$. The ratio is $9/8$: B has 12.5% more variance, not 12.5% more SD.

5. Bias and consistency (medium)

Correct answer: E. A is unbiased and inconsistent; B is biased and consistent; C is biased and inconsistent.

$E(A_n) = \mu$ but its nondegenerate distribution never changes. $E(B_n) = \mu + 3/n$, so it is biased for each finite n but converges in probability to μ by the LLN. $C_n \rightarrow_p \mu + 3$ and has constant bias 3. Nonvanishing variance alone is not a general proof of inconsistency; the unchanged distribution proves it for A.

6. Sample variance and standard error (medium)

Correct answer: A. $(\frac{20}{3}, \frac{5}{3})$

The sample mean is 7; squared deviations total $9 + 1 + 1 + 9 = 20$. Hence $s_Y^2 = 20/3$ and $SE^2 = s_Y^2/4 = 5/3$. These calculations do not justify an exact Student t reference without normality or a normal approximation with just four observations.

7. A two-sided test of a mean (medium)

Correct answer: A. $t = 2$, $p \approx 0.0455$; reject.

$SE = 12/\sqrt{144} = 1$; $t = (23 - 21)/1 = 2$. Both tails count: $p \approx 2[1 - \Phi(2)] = 0.0455 < 0.05$, so reject. Finite positive variance supplies the usual CLT/Slutsky conditions for the approximation. A p-value is not the probability the null is true. At 1% the same result would not reject.

8. OLS coefficients from summaries (medium)**Correct answer: B.** (4, 3)

$\hat{\beta}_1 = S_{XY}/S_{XX} = 3$ and $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1\bar{X} = 13 - 3(3) = 4$. The line passes through the means. These algebraic facts need no causal assumption.

9. Fit and residual scale (medium)**Correct answer: D.** ($\frac{3}{4}$, 3)

$R^2 = 1 - SSR/TSS = 1 - 1/4 = 3/4$. $SER = \sqrt{SSR/(n-2)} = \sqrt{144/16} = 3$ in outcome units. R-squared is unitless and establishes neither causality nor out-of-sample accuracy; SER is not the slope's SE.

10. Omission and the probability limit (medium)**Correct answer: G.** (3, 3)

The auxiliary slope is $\pi = -2/4 = -1/2$. The distortion is $2\pi = -1$, so $b_s = 4 - 1 = 3$. The LLN sends the sample covariance and variance to their population counterparts, hence $\hat{b}_s \rightarrow_p b_s = 3$, not the causal target 4. More observations do not remove the omitted pathway.

11. Missing joint cells and reverse conditioning (hard)**Correct answer: C.** ($\frac{7}{20}$, $\frac{1}{4}$, $\frac{5}{7}$)

Since Y is 0 or 4, $\Pr(Y = 4) = E(Y)/4 = 0.35$. Thus $b = 0.35 - 0.10 = 0.25$, and $a = 0.60 - b = 0.35$. Reverse conditioning divides by the Y marginal: $\Pr(X = 1 \mid Y = 4) = b/\Pr(Y = 4) = \frac{5}{7}$. Dividing by 0.60 answers the opposite conditioning question.

12. Repair an iterated-expectations proof (hard)**Correct answer: E.** (p_x, p_{xy}); independence is unnecessary.

The outer expectation weights each X group by p_x . The multiplication rule gives $\Pr(Y = y \mid X = x)p_x = (p_{xy}/p_x)p_x = p_{xy}$ for each pair. Then $\sum_x p_{xy} = p_y$, completing the proof. Independence is not used. A product of marginals may give the same final summed expectation, but it is not a valid termwise replacement for a dependent joint distribution.

13. Zero covariance with a nonlinear error (hard)**Correct answer: D.** (0 , $\frac{49}{4}$); zero conditional mean fails.

Symmetry gives $E(X) = E(X^3) = 0$, so $E(Xu) = 0$; also $E(u) = 0$ by centering. But $E(X^2u) = E(X^4) - [E(X^2)]^2 = \text{Var}(X^2) = (4^2 - 3^2)^2/4 = \frac{49}{4} > 0$. Since u is determined by X , $E(u \mid X) = X^2 - \frac{25}{2}$ takes the nonzero values $\pm \frac{7}{2}$. Zero covariance does not imply zero conditional mean or independence.

14. Weights, concentration, and relative variance (hard)**Correct answer: F.** ($\frac{1}{9}$, 10)

Y_1 and $\bar{Y}_{-1}$ are independent, with variances σ^2 and $\sigma^2/(n-1)$. Hence $\text{Var}(T_n)/\sigma^2 = w^2 + (1-w)^2/(n-1) \rightarrow w^2 = \frac{1}{9}$. For S, divide $\sigma^2[k^2/n + (1-k/\sqrt{n})^2/(n-1)]$ by σ^2/n : $k^2 + \frac{n}{n-1}(1-k/\sqrt{n})^2 \rightarrow k^2 + 1 = 10$. Both are unbiased. T converges in probability to $wY_1 + (1-w)\mu$, so is inconsistent; S is consistent but pays a first-order variance cost.

15. Outcome-dependent weights (hard)

Correct answer: C. 0.51; step (2), because weights depend on outcomes.

This rule selects $T = \max(Y_1, Y_2)$, which is zero only if both outcomes are zero. Thus $E(T) = 1 - (1 - p)^2 = 2p - p^2 = 0.51$, with upward bias $p(1 - p) = 0.21$. Step (1) is linearity and is valid without independence. Step (2) incorrectly factors dependent products: independence between the Y observations does not make a selected weight independent of its own Y . Step (3) is valid algebra on the expression immediately before it.

16. Vanishing bias at the CLT scale (hard)

Correct answer: C. $(\mu, N(-2, 1))$

The LLN and vanishing deterministic shift give $B_n \rightarrow_p \mu$. Yet $\sqrt{n}(B_n - \mu)/\sigma = \sqrt{n}(\bar{Y}_n - \mu)/\sigma - 2 \rightarrow_d N(-2, 1)$. The finite-sample bias is $-2\sigma/\sqrt{n}$: it vanishes in levels but remains 2 sampling SDs below zero. A deterministic shift does not add variance.

17. Duplicated observations and testing (hard)

Correct answer: D. $z = 2$, $p \approx 0.0455$; do not reject.

The copied mean equals $4\sum_i Y_i/(4n) = \bar{Y}$ exactly. All four copies of each observation are perfectly dependent, so its sampling SD stays $12/\sqrt{144} = 1$. Therefore $z = (13 - 11)/1 = 2$ and $p = 2(1 - 0.97725) = 0.0455 > 0.01$: do not reject. Treating rows as independent would incorrectly halve the SE and report $z=4$. The normal reference is exact here under the stated normal, known-variance model; CDF values are rounded.

18. Recover outcomes and fit from normal equations (hard)

Correct answer: A. $(4, 7, \frac{5}{9})$

The endpoint residuals are both 3. Write $\hat{u} = (3, a_* - (7), b_* - (10), 3)$. The two normal equations imply $a_* + b_* = 11$ and $a_* + 2b_* = 18$. Subtraction gives $b_* = 7$ and $a_* = 4$; residuals are $3(1, -1, -1, 1)$. Fitted deviations are $3(-3/2, -1/2, 1/2, 3/2)$, so $ESS = 5(3^2)$ and $SSR = 4(3^2)$. By the intercept-OLS decomposition $TSS = 9(3^2)$, hence $R^2 = 5/9$. Missing outcomes need not equal fitted values.

19. Complete the fit-decomposition argument (hard)

Correct answer: F. $\hat{\beta}_1[\sum_i X_i \hat{u}_i - \bar{X} \sum_i \hat{u}_i]$; both sample normal equations.

The intercept equation gives $\hat{Y}_i - \bar{Y} = \hat{\beta}_1(X_i - \bar{X})$. Substitution yields $C = \hat{\beta}_1[\sum_i X_i \hat{u}_i - \bar{X} \sum_i \hat{u}_i]$. Both sums vanish by the fitted normal equations. Thus $TSS = ESS + SSR$. No population exogeneity, independence, normality, or causal interpretation is needed for this sample algebra. Option using only $\hat{\beta}_1 \sum X_i \hat{u}_i$ has the same zero value at the fit, but its claim that population zero conditional mean is required is false.

20. Confounding and concentration around the wrong target (hard)

Correct answer: G. $(\frac{3}{2}, \approx 0.1573, 0)$

$\text{Cov}(X, Z) = (1/2)(2)(2) = 2$, so $\pi = 2/4 = 1/2$ and the omitted distortion is $1/2$. Hence $\beta = b_s - 1/2 = \frac{3}{2}$. At $n=400$ the approximate sampling SD is $5/20 = 1/4$. Relative to the causal target, $\hat{b}_s - \beta$ is approximately $N(1/2, 1/16)$. Bounds $[-1/4, 1/4]$ standardize to $[-3, -1]$: probability $\approx \Phi(-1) - \Phi(-3) = \Phi(3) - \Phi(1) = 0.15731$. The LLN sends $\hat{b}_s$ to $b_s = \beta + 1/2$, outside this fixed tolerance, so the probability tends to zero. The CLT describes shrinking variation around the short slope, not the causal slope.

Form M8VN: worked solutions

1. Conditioning and group averages (medium)

Correct answer: C. $(\frac{8}{3}, 2)$

$\Pr(X = 1) = 0.60$, so $E(Y | X = 1) = 4(0.40/0.60) = \frac{8}{3}$. Marginally $\Pr(Y = 4) = 0.50$, hence $E(Y) = 4(0.50) = 2$. Equivalently, average the two conditional means with weights 0.40 and 0.60, not equal weights. These are descriptive population quantities.

2. Moments after a change of units (medium)

Correct answer: B. $(5, 32)$

$E(Y) = 2$ and $E(Y^2) = 6$, so $\text{Var}(Y) = 6 - 2^2 = 2$. Therefore $E(W) = 13 - 4(2) = 5$ and $\text{Var}(W) = (-4)^2(2) = 32$. The intercept shifts the mean only; variance uses the squared scale.

3. Rescaling covariance and correlation (medium)

Correct answer: A. $(36, \frac{1}{2})$

$\text{Cov}(X, Y) = (-1/2)(2)(6) = -6$. Thus $\text{Cov}(W, V) = 2(-3)(-6) = 36$. The single negative scale reverses the correlation sign, giving $+1/2$; location shifts do not affect covariance or correlation.

4. Efficiency of fixed weights (medium)

Correct answer: G. $(6, \frac{9}{8})$

Both sets of fixed weights sum to one, so both rules are unbiased. Independence and squared weights give $\text{Var}(B) = 16(1/4 + 1/16 + 1/16) = 6$, while $\text{Var}(A) = \frac{16}{3}$. The ratio is $9/8$: B has 12.5% more variance, not 12.5% more SD.

5. Bias and consistency (medium)

Correct answer: F. A is unbiased and inconsistent; B is biased and consistent; C is biased and inconsistent.

$E(A_n) = \mu$ but its nondegenerate distribution never changes. $E(B_n) = \mu + 4/n$, so it is biased for each finite n but converges in probability to μ by the LLN. $C_n \rightarrow_p \mu + 4$ and has constant bias 4. Nonvanishing variance alone is not a general proof of inconsistency; the unchanged distribution proves it for A.

6. Sample variance and standard error (medium)

Correct answer: B. $(\frac{20}{3}, \frac{5}{3})$

The sample mean is 8; squared deviations total $9 + 1 + 1 + 9 = 20$. Hence $s_Y^2 = 20/3$ and $SE^2 = s_Y^2/4 = 5/3$. These calculations do not justify an exact Student t reference without normality or a normal approximation with just four observations.

7. A two-sided test of a mean (medium)

Correct answer: E. $t = 2$, $p \approx 0.0455$; reject.

$SE = 12/\sqrt{144} = 1$; $t = (24 - 22)/1 = 2$. Both tails count: $p \approx 2[1 - \Phi(2)] = 0.0455 < 0.05$, so reject. Finite positive variance supplies the usual CLT/Slutsky conditions for the approximation. A p-value is not the probability the null is true. At 1% the same result would not reject.

8. OLS coefficients from summaries (medium)

Correct answer: F. (5, 4)

$\hat{\beta}_1 = S_{XY}/S_{XX} = 4$ and $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1\bar{X} = 17 - 4(3) = 5$. The line passes through the means. These algebraic facts need no causal assumption.

9. Fit and residual scale (medium)

Correct answer: G. ($\frac{3}{4}$, 4)

$R^2 = 1 - SSR/TSS = 1 - 1/4 = 3/4$. $SER = \sqrt{SSR/(n-2)} = \sqrt{256/16} = 4$ in outcome units. R-squared is unitless and establishes neither causality nor out-of-sample accuracy; SER is not the slope's SE.

10. Omission and the probability limit (medium)

Correct answer: D. (4, 4)

The auxiliary slope is $\pi = -2/4 = -1/2$. The distortion is $2\pi = -1$, so $b_s = 5 - 1 = 4$. The LLN sends the sample covariance and variance to their population counterparts, hence $\hat{b}_s \rightarrow_p b_s = 4$, not the causal target 5. More observations do not remove the omitted pathway.

11. Missing joint cells and reverse conditioning (hard)

Correct answer: E. ($\frac{3}{10}$, $\frac{3}{10}$, $\frac{3}{4}$)

Since Y is 0 or 5, $\Pr(Y = 5) = E(Y)/5 = 0.40$. Thus $b = 0.40 - 0.10 = 0.30$, and $a = 0.60 - b = 0.30$. Reverse conditioning divides by the Y marginal: $\Pr(X = 1 \mid Y = 5) = b/\Pr(Y = 5) = \frac{3}{4}$. Dividing by 0.60 answers the opposite conditioning question.

12. Repair an iterated-expectations proof (hard)

Correct answer: C. (p_x, p_{xy}); independence is unnecessary.

The outer expectation weights each X group by p_x . The multiplication rule gives $\Pr(Y = y \mid X = x)p_x = (p_{xy}/p_x)p_x = p_{xy}$ for each pair. Then $\sum_x p_{xy} = p_y$, completing the proof. Independence is not used. A product of marginals may give the same final summed expectation, but it is not a valid termwise replacement for a dependent joint distribution.

13. Zero covariance with a nonlinear error (hard)

Correct answer: C. (0, $\frac{81}{4}$); zero conditional mean fails.

Symmetry gives $E(X) = E(X^3) = 0$, so $E(Xu) = 0$; also $E(u) = 0$ by centering. But $E(X^2u) = E(X^4) - [E(X^2)]^2 = \text{Var}(X^2) = (5^2 - 4^2)^2/4 = \frac{81}{4} > 0$. Since u is determined by X , $E(u \mid X) = X^2 - \frac{41}{2}$ takes the nonzero values $\pm \frac{9}{2}$. Zero covariance does not imply zero conditional mean or independence.

14. Weights, concentration, and relative variance (hard)

Correct answer: D. ($\frac{4}{9}$, 5)

Y_1 and $\bar{Y}_{-1}$ are independent, with variances σ^2 and $\sigma^2/(n-1)$. Hence $\text{Var}(T_n)/\sigma^2 = w^2 + (1-w)^2/(n-1) \rightarrow w^2 = \frac{4}{9}$. For S, divide $\sigma^2[k^2/n + (1-k/\sqrt{n})^2/(n-1)]$ by σ^2/n : $k^2 + \frac{n}{n-1}(1-k/\sqrt{n})^2 \rightarrow k^2 + 1 = 5$. Both are unbiased. T converges in probability to $wY_1 + (1-w)\mu$, so is inconsistent; S is consistent but pays a first-order variance cost.

15. Outcome-dependent weights (hard)

Correct answer: F. 0.64; step (2), because weights depend on outcomes.

This rule selects $T = \max(Y_1, Y_2)$, which is zero only if both outcomes are zero. Thus $E(T) = 1 - (1 - p)^2 = 2p - p^2 = 0.64$, with upward bias $p(1 - p) = 0.24$. Step (1) is linearity and is valid without independence. Step (2) incorrectly factors dependent products: independence between the Y observations does not make a selected weight independent of its own Y . Step (3) is valid algebra on the expression immediately before it.

16. Vanishing bias at the CLT scale (hard)

Correct answer: A. $(\mu, N(-3, 1))$

The LLN and vanishing deterministic shift give $B_n \rightarrow_p \mu$. Yet $\sqrt{n}(B_n - \mu)/\sigma = \sqrt{n}(\bar{Y}_n - \mu)/\sigma - 3 \rightarrow_d N(-3, 1)$. The finite-sample bias is $-3\sigma/\sqrt{n}$: it vanishes in levels but remains 3 sampling SDs below zero. A deterministic shift does not add variance.

17. Duplicated observations and testing (hard)

Correct answer: G. $z = 2$, $p \approx 0.0455$; do not reject.

The copied mean equals $4\sum_i Y_i/(4n) = \bar{Y}$ exactly. All four copies of each observation are perfectly dependent, so its sampling SD stays $14/\sqrt{196} = 1$. Therefore $z = (14 - 12)/1 = 2$ and $p = 2(1 - 0.97725) = 0.0455 > 0.01$: do not reject. Treating rows as independent would incorrectly halve the SE and report $z=4$. The normal reference is exact here under the stated normal, known-variance model; CDF values are rounded.

18. Recover outcomes and fit from normal equations (hard)

Correct answer: E. $(5, 9, \frac{5}{9})$

The endpoint residuals are both 4. Write $\hat{u} = (4, a_* - (9), b_* - (13), 4)$. The two normal equations imply $a_* + b_* = 14$ and $a_* + 2b_* = 23$. Subtraction gives $b_* = 9$ and $a_* = 5$; residuals are $4(1, -1, -1, 1)$. Fitted deviations are $4(-3/2, -1/2, 1/2, 3/2)$, so $ESS = 5(4^2)$ and $SSR = 4(4^2)$. By the intercept-OLS decomposition $TSS = 9(4^2)$, hence $R^2 = 5/9$. Missing outcomes need not equal fitted values.

19. Complete the fit-decomposition argument (hard)

Correct answer: B. $\hat{\beta}_1[\sum_i X_i \hat{u}_i - \bar{X} \sum_i \hat{u}_i]$; both sample normal equations.

The intercept equation gives $\hat{Y}_i - \bar{Y} = \hat{\beta}_1(X_i - \bar{X})$. Substitution yields $C = \hat{\beta}_1[\sum_i X_i \hat{u}_i - \bar{X} \sum_i \hat{u}_i]$. Both sums vanish by the fitted normal equations. Thus $TSS = ESS + SSR$. No population exogeneity, independence, normality, or causal interpretation is needed for this sample algebra. Option using only $\hat{\beta}_1 \sum X_i \hat{u}_i$ has the same zero value at the fit, but its claim that population zero conditional mean is required is false.

20. Confounding and concentration around the wrong target (hard)

Correct answer: A. $(\frac{5}{2}, \approx 0.1573, 0)$

$\text{Cov}(X, Z) = (1/2)(2)(2) = 2$, so $\pi = 2/4 = 1/2$ and the omitted distortion is $1/2$. Hence $\beta = b_s - 1/2 = \frac{5}{2}$. At $n=400$ the approximate sampling SD is $5/20 = 1/4$. Relative to the causal target, $\hat{b}_s - \beta$ is approximately $N(1/2, 1/16)$. Bounds $[-1/4, 1/4]$ standardize to $[-3, -1]$: probability $\approx \Phi(-1) - \Phi(-3) = \Phi(3) - \Phi(1) = 0.15731$. The LLN sends $\hat{b}_s$ to $b_s = \beta + 1/2$, outside this fixed tolerance, so the probability tends to zero. The CLT describes shrinking variation around the short slope, not the causal slope.

Form M2HF: worked solutions

1. Conditioning and group averages (medium)

Correct answer: F. $(\frac{10}{3}, \frac{5}{2})$

$\Pr(X = 1) = 0.60$, so $E(Y | X = 1) = 5(0.40/0.60) = \frac{10}{3}$. Marginally $\Pr(Y = 5) = 0.50$, hence $E(Y) = 5(0.50) = \frac{5}{2}$. Equivalently, average the two conditional means with weights 0.40 and 0.60, not equal weights. These are descriptive population quantities.

2. Moments after a change of units (medium)

Correct answer: D. (6, 50)

$E(Y) = 2$ and $E(Y^2) = 6$, so $\text{Var}(Y) = 6 - 2^2 = 2$. Therefore $E(W) = 16 - 5(2) = 6$ and $\text{Var}(W) = (-5)^2(2) = 50$. The intercept shifts the mean only; variance uses the squared scale.

3. Rescaling covariance and correlation (medium)

Correct answer: E. $(42, \frac{1}{2})$

$\text{Cov}(X, Y) = (-1/2)(2)(7) = -7$. Thus $\text{Cov}(W, V) = 2(-3)(-7) = 42$. The single negative scale reverses the correlation sign, giving $+1/2$; location shifts do not affect covariance or correlation.

4. Efficiency of fixed weights (medium)

Correct answer: A. $(\frac{75}{8}, \frac{9}{8})$

Both sets of fixed weights sum to one, so both rules are unbiased. Independence and squared weights give $\text{Var}(B) = 25(1/4 + 1/16 + 1/16) = \frac{75}{8}$, while $\text{Var}(A) = \frac{25}{3}$. The ratio is $9/8$: B has 12.5% more variance, not 12.5% more SD.

5. Bias and consistency (medium)

Correct answer: G. A is unbiased and inconsistent; B is biased and consistent; C is biased and inconsistent.

$E(A_n) = \mu$ but its nondegenerate distribution never changes. $E(B_n) = \mu + 5/n$, so it is biased for each finite n but converges in probability to μ by the LLN. $C_n \rightarrow_p \mu + 5$ and has constant bias 5. Nonvanishing variance alone is not a general proof of inconsistency; the unchanged distribution proves it for A.

6. Sample variance and standard error (medium)

Correct answer: D. $(\frac{20}{3}, \frac{5}{3})$

The sample mean is 9; squared deviations total $9 + 1 + 1 + 9 = 20$. Hence $s_Y^2 = 20/3$ and $SE^2 = s_Y^2/4 = 5/3$. These calculations do not justify an exact Student t reference without normality or a normal approximation with just four observations.

7. A two-sided test of a mean (medium)

Correct answer: B. $t = 2$, $p \approx 0.0455$; reject.

$SE = 12/\sqrt{144} = 1$; $t = (25 - 23)/1 = 2$. Both tails count: $p \approx 2[1 - \Phi(2)] = 0.0455 < 0.05$, so reject. Finite positive variance supplies the usual CLT/Slutsky conditions for the approximation. A p-value is not the probability the null is true. At 1% the same result would not reject.

8. OLS coefficients from summaries (medium)**Correct answer: A.** (6, 5)

$\hat{\beta}_1 = S_{XY}/S_{XX} = 5$ and $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1\bar{X} = 21 - 5(3) = 6$. The line passes through the means. These algebraic facts need no causal assumption.

9. Fit and residual scale (medium)**Correct answer: E.** ($\frac{3}{4}$, 5)

$R^2 = 1 - SSR/TSS = 1 - 1/4 = 3/4$. $SER = \sqrt{SSR/(n-2)} = \sqrt{400/16} = 5$ in outcome units. R-squared is unitless and establishes neither causality nor out-of-sample accuracy; SER is not the slope's SE.

10. Omission and the probability limit (medium)**Correct answer: A.** (5, 5)

The auxiliary slope is $\pi = -2/4 = -1/2$. The distortion is $2\pi = -1$, so $b_s = 6 - 1 = 5$. The LLN sends the sample covariance and variance to their population counterparts, hence $\hat{b}_s \rightarrow_p b_s = 5$, not the causal target 6. More observations do not remove the omitted pathway.

11. Missing joint cells and reverse conditioning (hard)**Correct answer: B.** ($\frac{1}{4}$, $\frac{7}{20}$, $\frac{7}{9}$)

Since Y is 0 or 6, $\Pr(Y = 6) = E(Y)/6 = 0.45$. Thus $b = 0.45 - 0.10 = 0.35$, and $a = 0.60 - b = 0.25$. Reverse conditioning divides by the Y marginal: $\Pr(X = 1 \mid Y = 6) = b/\Pr(Y = 6) = \frac{7}{9}$. Dividing by 0.60 answers the opposite conditioning question.

12. Repair an iterated-expectations proof (hard)**Correct answer: G.** (p_x, p_{xy}); independence is unnecessary.

The outer expectation weights each X group by p_x . The multiplication rule gives $\Pr(Y = y \mid X = x)p_x = (p_{xy}/p_x)p_x = p_{xy}$ for each pair. Then $\sum_x p_{xy} = p_y$, completing the proof. Independence is not used. A product of marginals may give the same final summed expectation, but it is not a valid termwise replacement for a dependent joint distribution.

13. Zero covariance with a nonlinear error (hard)**Correct answer: E.** (0 , $\frac{121}{4}$); zero conditional mean fails.

Symmetry gives $E(X) = E(X^3) = 0$, so $E(Xu) = 0$; also $E(u) = 0$ by centering. But $E(X^2u) = E(X^4) - [E(X^2)]^2 = \text{Var}(X^2) = (6^2 - 5^2)^2/4 = \frac{121}{4} > 0$. Since u is determined by X , $E(u \mid X) = X^2 - \frac{61}{2}$ takes the nonzero values $\pm \frac{11}{2}$. Zero covariance does not imply zero conditional mean or independence.

14. Weights, concentration, and relative variance (hard)**Correct answer: G.** ($\frac{1}{16}$, 10)

Y_1 and $\bar{Y}_{-1}$ are independent, with variances σ^2 and $\sigma^2/(n-1)$. Hence $\text{Var}(T_n)/\sigma^2 = w^2 + (1-w)^2/(n-1) \rightarrow w^2 = \frac{1}{16}$. For S, divide $\sigma^2[k^2/n + (1-k/\sqrt{n})^2/(n-1)]$ by σ^2/n : $k^2 + \frac{n}{n-1}(1-k/\sqrt{n})^2 \rightarrow k^2 + 1 = 10$. Both are unbiased. T converges in probability to $wY_1 + (1-w)\mu$, so is inconsistent; S is consistent but pays a first-order variance cost.

15. Outcome-dependent weights (hard)

Correct answer: B. 0.84; step (2), because weights depend on outcomes.

This rule selects $T = \max(Y_1, Y_2)$, which is zero only if both outcomes are zero. Thus $E(T) = 1 - (1 - p)^2 = 2p - p^2 = 0.84$, with upward bias $p(1 - p) = 0.24$. Step (1) is linearity and is valid without independence. Step (2) incorrectly factors dependent products: independence between the Y observations does not make a selected weight independent of its own Y . Step (3) is valid algebra on the expression immediately before it.

16. Vanishing bias at the CLT scale (hard)

Correct answer: F. $(\mu, N(-4, 1))$

The LLN and vanishing deterministic shift give $B_n \rightarrow_p \mu$. Yet $\sqrt{n}(B_n - \mu)/\sigma = \sqrt{n}(\bar{Y}_n - \mu)/\sigma - 4 \rightarrow_d N(-4, 1)$. The finite-sample bias is $-4\sigma/\sqrt{n}$: it vanishes in levels but remains 4 sampling SDs below zero. A deterministic shift does not add variance.

17. Duplicated observations and testing (hard)

Correct answer: C. $z = 2$, $p \approx 0.0455$; do not reject.

The copied mean equals $4\sum_i Y_i/(4n) = \bar{Y}$ exactly. All four copies of each observation are perfectly dependent, so its sampling SD stays $16/\sqrt{256} = 1$. Therefore $z = (15 - 13)/1 = 2$ and $p = 2(1 - 0.97725) = 0.0455 > 0.01$: do not reject. Treating rows as independent would incorrectly halve the SE and report $z=4$. The normal reference is exact here under the stated normal, known-variance model; CDF values are rounded.

18. Recover outcomes and fit from normal equations (hard)

Correct answer: C. $(6, 11, \frac{5}{9})$

The endpoint residuals are both 5. Write $\hat{u} = (5, a_* - (11), b_* - (16), 5)$. The two normal equations imply $a_* + b_* = 17$ and $a_* + 2b_* = 28$. Subtraction gives $b_* = 11$ and $a_* = 6$; residuals are $5(1, -1, -1, 1)$. Fitted deviations are $5(-3/2, -1/2, 1/2, 3/2)$, so $ESS = 5(5^2)$ and $SSR = 4(5^2)$. By the intercept-OLS decomposition $TSS = 9(5^2)$, hence $R^2 = 5/9$. Missing outcomes need not equal fitted values.

19. Complete the fit-decomposition argument (hard)

Correct answer: D. $\hat{\beta}_1[\sum_i X_i \hat{u}_i - \bar{X} \sum_i \hat{u}_i]$; both sample normal equations.

The intercept equation gives $\hat{Y}_i - \bar{Y} = \hat{\beta}_1(X_i - \bar{X})$. Substitution yields $C = \hat{\beta}_1[\sum_i X_i \hat{u}_i - \bar{X} \sum_i \hat{u}_i]$. Both sums vanish by the fitted normal equations. Thus $TSS = ESS + SSR$. No population exogeneity, independence, normality, or causal interpretation is needed for this sample algebra. Option using only $\hat{\beta}_1 \sum X_i \hat{u}_i$ has the same zero value at the fit, but its claim that population zero conditional mean is required is false.

20. Confounding and concentration around the wrong target (hard)

Correct answer: F. $(\frac{7}{2}, \approx 0.1573, 0)$

$\text{Cov}(X, Z) = (1/2)(2)(2) = 2$, so $\pi = 2/4 = 1/2$ and the omitted distortion is $1/2$. Hence $\beta = b_s - 1/2 = \frac{7}{2}$. At $n=400$ the approximate sampling SD is $5/20 = 1/4$. Relative to the causal target, $\hat{b}_s - \beta$ is approximately $N(1/2, 1/16)$. Bounds $[-1/4, 1/4]$ standardize to $[-3, -1]$: probability $\approx \Phi(-1) - \Phi(-3) = \Phi(3) - \Phi(1) = 0.15731$. The LLN sends $\hat{b}_s$ to $b_s = \beta + 1/2$, outside this fixed tolerance, so the probability tends to zero. The CLT describes shrinking variation around the short slope, not the causal slope.

Form M6WP: worked solutions

1. Conditioning and group averages (medium)

Correct answer: B. (4, 3)

$\Pr(X = 1) = 0.60$, so $E(Y | X = 1) = 6(0.40/0.60) = 4$. Marginally $\Pr(Y = 6) = 0.50$, hence $E(Y) = 6(0.50) = 3$. Equivalently, average the two conditional means with weights 0.40 and 0.60, not equal weights. These are descriptive population quantities.

2. Moments after a change of units (medium)

Correct answer: A. (7, 72)

$E(Y) = 2$ and $E(Y^2) = 6$, so $\text{Var}(Y) = 6 - 2^2 = 2$. Therefore $E(W) = 19 - 6(2) = 7$ and $\text{Var}(W) = (-6)^2(2) = 72$. The intercept shifts the mean only; variance uses the squared scale.

3. Rescaling covariance and correlation (medium)

Correct answer: G. (48, $\frac{1}{2}$)

$\text{Cov}(X, Y) = (-1/2)(2)(8) = -8$. Thus $\text{Cov}(W, V) = 2(-3)(-8) = 48$. The single negative scale reverses the correlation sign, giving $+1/2$; location shifts do not affect covariance or correlation.

4. Efficiency of fixed weights (medium)

Correct answer: E. ($\frac{27}{2}$, $\frac{9}{8}$)

Both sets of fixed weights sum to one, so both rules are unbiased. Independence and squared weights give $\text{Var}(B) = 36(1/4 + 1/16 + 1/16) = \frac{27}{2}$, while $\text{Var}(A) = 12$. The ratio is $9/8$: B has 12.5% more variance, not 12.5% more SD.

5. Bias and consistency (medium)

Correct answer: D. A is unbiased and inconsistent; B is biased and consistent; C is biased and inconsistent.

$E(A_n) = \mu$ but its nondegenerate distribution never changes. $E(B_n) = \mu + 6/n$, so it is biased for each finite n but converges in probability to μ by the LLN. $C_n \rightarrow_p \mu + 6$ and has constant bias 6. Nonvanishing variance alone is not a general proof of inconsistency; the unchanged distribution proves it for A.

6. Sample variance and standard error (medium)

Correct answer: C. ($\frac{20}{3}$, $\frac{5}{3}$)

The sample mean is 10; squared deviations total $9 + 1 + 1 + 9 = 20$. Hence $s_Y^2 = 20/3$ and $SE^2 = s_Y^2/4 = 5/3$. These calculations do not justify an exact Student t reference without normality or a normal approximation with just four observations.

7. A two-sided test of a mean (medium)

Correct answer: F. $t = 2$, $p \approx 0.0455$; reject.

$SE = 12/\sqrt{144} = 1$; $t = (26 - 24)/1 = 2$. Both tails count: $p \approx 2[1 - \Phi(2)] = 0.0455 < 0.05$, so reject. Finite positive variance supplies the usual CLT/Slutsky conditions for the approximation. A p-value is not the probability the null is true. At 1% the same result would not reject.

8. OLS coefficients from summaries (medium)**Correct answer: G.** (7, 6)

$\hat{\beta}_1 = S_{XY}/S_{XX} = 6$ and $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1\bar{X} = 25 - 6(3) = 7$. The line passes through the means. These algebraic facts need no causal assumption.

9. Fit and residual scale (medium)**Correct answer: C.** ($\frac{3}{4}$, 6)

$R^2 = 1 - SSR/TSS = 1 - 1/4 = 3/4$. $SER = \sqrt{SSR/(n-2)} = \sqrt{576/16} = 6$ in outcome units. R-squared is unitless and establishes neither causality nor out-of-sample accuracy; SER is not the slope's SE.

10. Omission and the probability limit (medium)**Correct answer: B.** (6, 6)

The auxiliary slope is $\pi = -2/4 = -1/2$. The distortion is $2\pi = -1$, so $b_s = 7 - 1 = 6$. The LLN sends the sample covariance and variance to their population counterparts, hence $\hat{b}_s \rightarrow_p b_s = 6$, not the causal target 7. More observations do not remove the omitted pathway.

11. Missing joint cells and reverse conditioning (hard)**Correct answer: D.** ($\frac{1}{5}$, $\frac{2}{5}$, $\frac{4}{5}$)

Since Y is 0 or 7, $\Pr(Y = 7) = E(Y)/7 = 0.50$. Thus $b = 0.50 - 0.10 = 0.40$, and $a = 0.60 - b = 0.20$. Reverse conditioning divides by the Y marginal: $\Pr(X = 1 \mid Y = 7) = b/\Pr(Y = 7) = \frac{4}{5}$. Dividing by 0.60 answers the opposite conditioning question.

12. Repair an iterated-expectations proof (hard)**Correct answer: F.** (p_x, p_{xy}); independence is unnecessary.

The outer expectation weights each X group by p_x . The multiplication rule gives $\Pr(Y = y \mid X = x)p_x = (p_{xy}/p_x)p_x = p_{xy}$ for each pair. Then $\sum_x p_{xy} = p_y$, completing the proof. Independence is not used. A product of marginals may give the same final summed expectation, but it is not a valid termwise replacement for a dependent joint distribution.

13. Zero covariance with a nonlinear error (hard)**Correct answer: A.** (0, $\frac{169}{4}$); zero conditional mean fails.

Symmetry gives $E(X) = E(X^3) = 0$, so $E(Xu) = 0$; also $E(u) = 0$ by centering. But $E(X^2u) = E(X^4) - [E(X^2)]^2 = \text{Var}(X^2) = (7^2 - 6^2)/4 = \frac{169}{4} > 0$. Since u is determined by X , $E(u \mid X) = X^2 - \frac{85}{2}$ takes the nonzero values $\pm \frac{13}{2}$. Zero covariance does not imply zero conditional mean or independence.

14. Weights, concentration, and relative variance (hard)**Correct answer: E.** ($\frac{9}{16}$, 5)

Y_1 and $\bar{Y}_{-1}$ are independent, with variances σ^2 and $\sigma^2/(n-1)$. Hence $\text{Var}(T_n)/\sigma^2 = w^2 + (1-w)^2/(n-1) \rightarrow w^2 = \frac{9}{16}$. For S, divide $\sigma^2[k^2/n + (1-k/\sqrt{n})^2/(n-1)]$ by σ^2/n : $k^2 + \frac{n}{n-1}(1-k/\sqrt{n})^2 \rightarrow k^2 + 1 = 5$. Both are unbiased. T converges in probability to $wY_1 + (1-w)\mu$, so is inconsistent; S is consistent but pays a first-order variance cost.

15. Outcome-dependent weights (hard)

Correct answer: A. 0.91; step (2), because weights depend on outcomes.

This rule selects $T = \max(Y_1, Y_2)$, which is zero only if both outcomes are zero. Thus $E(T) = 1 - (1 - p)^2 = 2p - p^2 = 0.91$, with upward bias $p(1 - p) = 0.21$. Step (1) is linearity and is valid without independence. Step (2) incorrectly factors dependent products: independence between the Y observations does not make a selected weight independent of its own Y . Step (3) is valid algebra on the expression immediately before it.

16. Vanishing bias at the CLT scale (hard)

Correct answer: B. $(\mu, N(-5, 1))$

The LLN and vanishing deterministic shift give $B_n \rightarrow_p \mu$. Yet $\sqrt{n}(B_n - \mu)/\sigma = \sqrt{n}(\bar{Y}_n - \mu)/\sigma - 5 \rightarrow_d N(-5, 1)$. The finite-sample bias is $-5\sigma/\sqrt{n}$: it vanishes in levels but remains 5 sampling SDs below zero. A deterministic shift does not add variance.

17. Duplicated observations and testing (hard)

Correct answer: E. $z = 2$, $p \approx 0.0455$; do not reject.

The copied mean equals $4\sum_i Y_i/(4n) = \bar{Y}$ exactly. All four copies of each observation are perfectly dependent, so its sampling SD stays $18/\sqrt{324} = 1$. Therefore $z = (16 - 14)/1 = 2$ and $p = 2(1 - 0.97725) = 0.0455 > 0.01$: do not reject. Treating rows as independent would incorrectly halve the SE and report $z=4$. The normal reference is exact here under the stated normal, known-variance model; CDF values are rounded.

18. Recover outcomes and fit from normal equations (hard)

Correct answer: F. $(7, 13, \frac{5}{9})$

The endpoint residuals are both 6. Write $\hat{u} = (6, a_* - (13), b_* - (19), 6)$. The two normal equations imply $a_* + b_* = 20$ and $a_* + 2b_* = 33$. Subtraction gives $b_* = 13$ and $a_* = 7$; residuals are $6(1, -1, -1, 1)$. Fitted deviations are $6(-3/2, -1/2, 1/2, 3/2)$, so $ESS = 5(6^2)$ and $SSR = 4(6^2)$. By the intercept-OLS decomposition $TSS = 9(6^2)$, hence $R^2 = 5/9$. Missing outcomes need not equal fitted values.

19. Complete the fit-decomposition argument (hard)

Correct answer: G. $\hat{\beta}_1[\sum_i X_i \hat{u}_i - \bar{X} \sum_i \hat{u}_i]$; both sample normal equations.

The intercept equation gives $\hat{Y}_i - \bar{Y} = \hat{\beta}_1(X_i - \bar{X})$. Substitution yields $C = \hat{\beta}_1[\sum_i X_i \hat{u}_i - \bar{X} \sum_i \hat{u}_i]$. Both sums vanish by the fitted normal equations. Thus $TSS = ESS + SSR$. No population exogeneity, independence, normality, or causal interpretation is needed for this sample algebra. Option using only $\hat{\beta}_1 \sum X_i \hat{u}_i$ has the same zero value at the fit, but its claim that population zero conditional mean is required is false.

20. Confounding and concentration around the wrong target (hard)

Correct answer: D. $(\frac{9}{2}, \approx 0.1573, 0)$

$\text{Cov}(X, Z) = (1/2)(2)(2) = 2$, so $\pi = 2/4 = 1/2$ and the omitted distortion is $1/2$. Hence $\beta = b_s - 1/2 = \frac{9}{2}$. At $n=400$ the approximate sampling SD is $5/20 = 1/4$. Relative to the causal target, $\hat{b}_s - \beta$ is approximately $N(1/2, 1/16)$. Bounds $[-1/4, 1/4]$ standardize to $[-3, -1]$: probability $\approx \Phi(-1) - \Phi(-3) = \Phi(3) - \Phi(1) = 0.15731$. The LLN sends $\hat{b}_s$ to $b_s = \beta + 1/2$, outside this fixed tolerance, so the probability tends to zero. The CLT describes shrinking variation around the short slope, not the causal slope.